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**Research and Development for  
Space Data System Standards**

**LOW DENSITY PARITY  
CHECK CODES FOR USE IN  
NEAR-EARTH AND DEEP  
SPACE APPLICATIONS**

**EXPERIMENTAL SPECIFICATION**

**CCSDS 131.1-O-1**

**August 2006**

## **AUTHORITY**

|           |                      |
|-----------|----------------------|
| Issue:    | Orange Book, Issue 1 |
| Date:     | August 2006          |
| Location: | Washington, DC, USA  |

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CCSDS Secretariat  
Office of Space Communication (Code M-3)  
National Aeronautics and Space Administration  
Washington, DC 20546, USA

## **FOREWORD**

This document is a CCSDS Experimental specification for a set of Low Density Parity Check (LDPC) codes. It was contributed to CCSDS by NASA.

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## **PREFACE**

This document is a CCSDS Experimental Specification. Its Experimental status indicates that it is part of a research or development effort based on prospective requirements, and as such it is not considered a Standards Track document. Experimental Specifications are intended to demonstrate technical feasibility in anticipation of a 'hard' requirement that has not yet emerged. Experimental work may be rapidly transferred onto the Standards Track should a hard requirement emerge in the future.

## **DOCUMENT CONTROL**

| <b>Document</b>    | <b>Title</b>                                                                                                                   | <b>Date</b>    | <b>Status</b>  |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------|----------------|
| CCSDS<br>131.1-O-1 | Low Density Parity Check Codes for<br>Use in Near-Earth and Deep Space<br>Applications, Experimental<br>Specification, Issue 1 | August<br>2006 | Original issue |

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## 1 INTRODUCTION

### 1.1 APPLICABILITY AND SCOPE

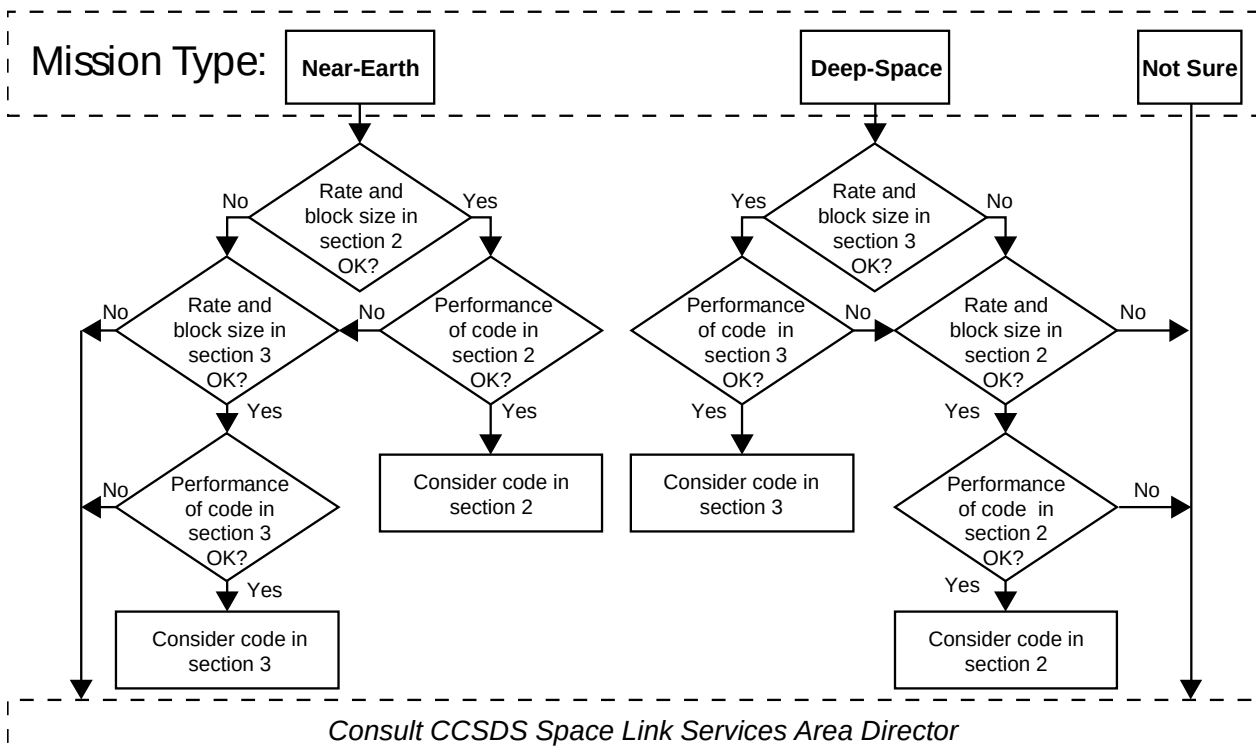
This CCSDS Experimental specification has been contributed to CCSDS by NASA. It describes a set of Low Density Parity Check (LDPC) codes including definition of their code structures, encoder implementations and experimental performance results.

This CCSDS Experimental specification is composed of two parts. The single rate 7/8 code described in section 2 was designed and optimized for the characteristics and requirements typical of many Near-Earth missions, whereas the set of codes described in section 3 was designed and optimized for the characteristics and requirements typical of many Deep-Space missions.

Users are cautioned that selection of the most appropriate code for a particular mission environment is a complex problem that involves evaluation of several characteristics, including:

- error rate performance, including error floors;
- encoder and decoder complexity, including the number of decoder iterations required to achieve desired performance;
- familial relationships between multiple codes;
- maturity and availability of implementations and implementation support;
- intellectual property issues;
- etc.

In addition, users may need to perform independent analysis and simulation. In order to begin the process of code selection, users are advised to approach the problem as follows:



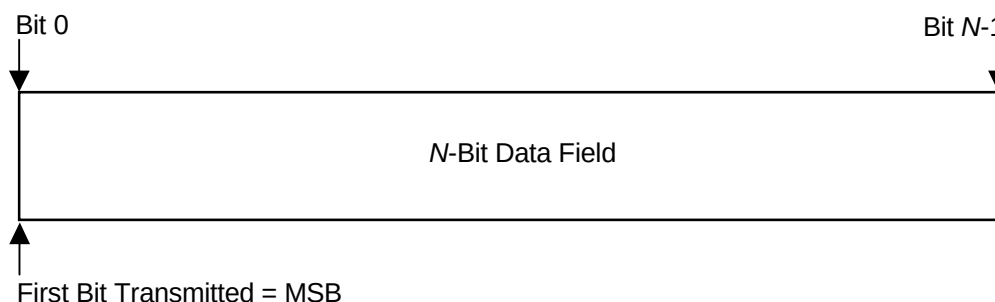
## 1.2 NOMENCLATURE

The following conventions apply throughout this Specification:

- a) the words 'shall' and 'must' imply a binding and verifiable specification;
- b) the word 'should' implies an optional, but desirable, specification;
- c) the word 'may' implies an optional specification;
- d) the words 'is', 'are', and 'will' imply statements of fact.

## 1.3 CONVENTIONS

In this document, the following convention is used to identify each bit in an  $N$ -bit field. The first bit in the field to be transmitted (i.e., the most left justified when drawing a figure) is defined to be 'Bit 0', the following bit is defined to be 'Bit 1', and so on up to 'Bit  $N-1$ '. When the field is used to express a binary value (such as a counter), the Most Significant Bit (MSB) shall be the first transmitted bit of the field, i.e., 'Bit 0' (see figure 1-1).



**Figure 1-1: Bit Numbering Convention**

The convention for matrices differs from that for bit fields. Matrices are indexed beginning with the number '1'.

In accordance with standard data-communications practice, data fields are often grouped into 8-bit 'words' which conform to the above convention. Throughout this Specification, such an 8-bit word is called an 'octet'.

The numbering for octets within a data structure starts with '0'.

## **1.4 REFERENCES**

The following document contains provisions which, through reference in this text, constitute provisions of this Experimental Specification. At the time of publication, the edition indicated was valid. All documents are subject to revision, and users of this Experimental Specification are encouraged to investigate the possibility of applying the most recent edition of the document indicated below. The CCSDS Secretariat maintains a register of currently valid CCSDS documents.

- [1] *TM Synchronization and Channel Coding*. Recommendation for Space Data System Standards, CCSDS 131.0-B-1. Blue Book. Issue 1. Washington, D.C.: CCSDS, September 2003.

NOTE – Annex C contains lists of informative references.

## 2 LOW DENSITY PARITY CHECK CODE OPTIMIZED FOR NEAR EARTH APPLICATIONS

### 2.1 OVERVIEW

#### 2.1.1 BACKGROUND

The mid-1990s were highlighted by the rediscovery of Low Density Parity Check codes (LDPCC) in the field of channel coding (reference [C1.1]). Originally invented by R. Gallager in his PhD thesis in 1961 (reference [C1.2]), this coding technique was largely forgotten for more than 30 years. The primary advance in LDPCC is the discovery of an iterative decoding algorithm, now called Belief Propagation (BP) decoding, which offers near-optimum performance for large linear LDPCC at a manageable complexity. LDPCC performance gains were difficult to realize technologically in the early 1960s. Several decades of VLSI development has finally made the implementation of these codes practical.

The original construction, now called Gallager LDPCC, has come to be regarded as a special class of LDPCC. Recent advances in LDPC code construction have resulted in the development of new codes with (arguably) improved performance over Gallager LDPCC. One class of these codes, irregular LDPCC (reference [C1.3]), demonstrates improved performance in the waterfall region. Disadvantages of irregular codes, however, include an increase, in general, in the number of iterations required for decoding convergence and an unequal error protection between code bits resulting from the irregular structure. Another class of LDPCC developed using algebraic construction based on finite geometries (reference [C1.4]) has been shown to provide very low error floors and very fast iterative convergence. These qualities make these codes a good fit for near Earth applications where very high data rates and high reliability are the driving requirements. A white paper based on Euclidean Geometry LDPCC (reference [C1.5]) was submitted to the CCSDS Channel Coding Subpanel P1B in the fall 2002 Houston meeting. That paper and subsequent research is the basis for the code presented in this Experimental Specification.

#### 2.1.2 TECHNICAL FOUNDATION

A linear block code is designated in this Experimental Specification by  $(n, k)$  where  $n$  is the length of the codeword (or block) and  $k$  is the length of the information sequence. LDPC codes are linear block codes in which the ratio of the total number of '1's to the total number of elements in the parity check matrix is  $\ll 0.5$ . The distribution of the '1's determine the structure and performance of the decoder. An LDPC code is defined by its parity check matrix. The  $k \times n$  generator matrix which is used to encode a linear block code can be derived from the parity check matrix through linear operations. (The reader is encouraged to review reference [C1.8] for an overview of linear block codes).

The LDPC code considered in this specification is a member of a class of codes called Quasi-Cyclic codes. The construction of these codes involves juxtaposing smaller circulants (or cyclic submatrices) to form a larger parity check or base matrix.

An example of a circulant is shown in figure 2-1. Notice that every row is one bit right cyclic shift (where the end bit is wrapped around to the beginning bit) of the previous row. The entire circulant is uniquely determined and specified by its first row. For this example the first row has four '1's or a row weight of four.

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

**Figure 2-1: Example of a 15 × 15 Circulant Matrix**

An example of a quasi-cyclic parity check matrix is shown in figure 2-2. In this case, a quasi-cyclic 10 × 25 matrix is formed by an array of 2 × 5 circulant submatrices of size 5 × 5. To unambiguously describe this matrix, only the position of the '1's in the first row of every circulant submatrix and the location of each submatrix within the base matrix is needed.

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

**Figure 2-2: Example of a Quasi-Cyclic Matrix**

Constructing parity check matrices in this manner produces two positive features:

- a) the encoding complexity can be made linear with the code length or parity bits using shift registers, and
- b) encoder and decoder routing complexity in the interconnections of integrated circuits is reduced.

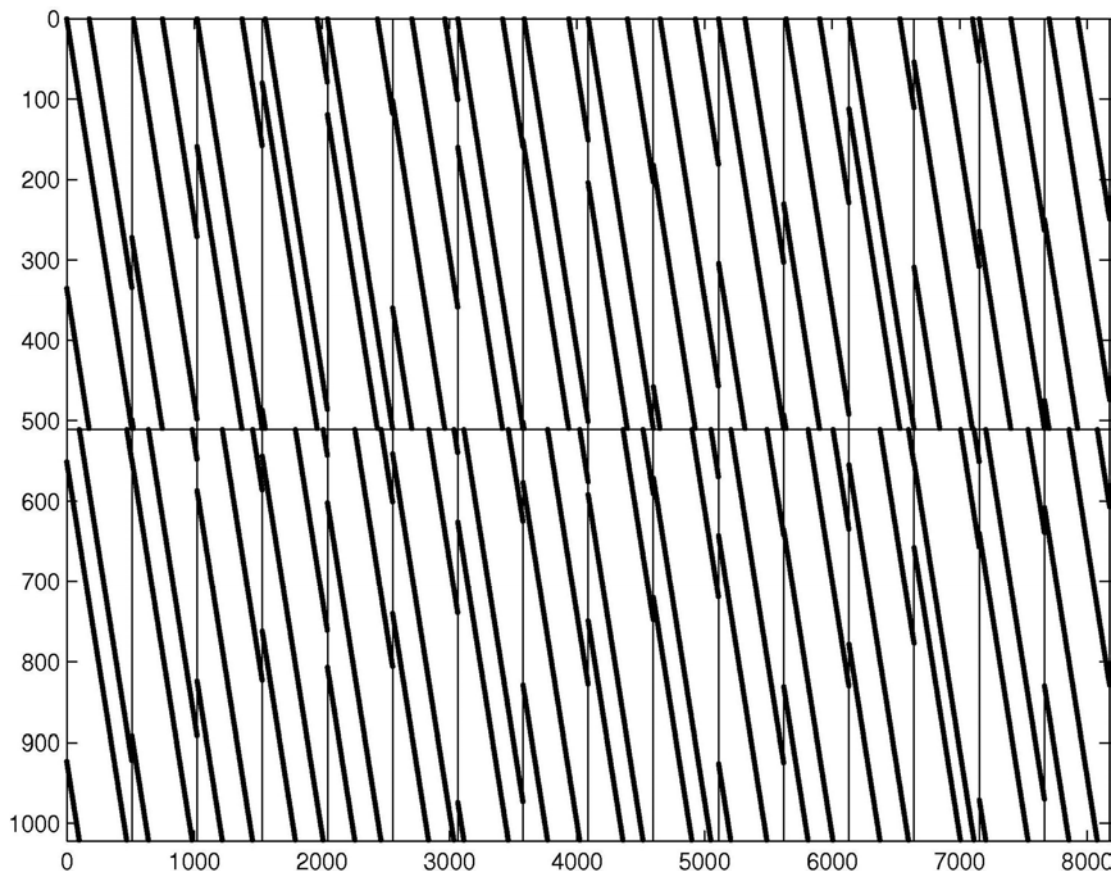
## 2.2 BASE (8176,7156) LDPC CODE

This section describes the base (8176, 7156) LDPC code. For reasons outlined below, implementations should shorten the base code according to the format described in subsection 2.5. The parity check matrix for the (8176, 7156) LDPC code is formed by using a  $2 \times 16$  array of  $511 \times 511$  square circulants. This creates a parity check matrix of dimension  $1022 \times 8176$ . The structure of the parity check base matrix is shown in figure 2-3.

$$\begin{bmatrix} A_{1,1} & A_{1,2} & A_{1,3} & A_{1,4} & A_{1,5} & A_{1,6} & A_{1,7} & A_{1,8} & A_{1,9} & A_{1,10} & A_{1,11} & A_{1,12} & A_{1,13} & A_{1,14} & A_{1,15} & A_{1,16} \\ A_{2,1} & A_{2,2} & A_{2,3} & A_{2,4} & A_{2,5} & A_{2,6} & A_{2,7} & A_{2,8} & A_{2,9} & A_{2,10} & A_{2,11} & A_{2,12} & A_{2,13} & A_{2,14} & A_{2,15} & A_{2,16} \end{bmatrix}$$

**Figure 2-3: Base Parity Check Matrix of the (8176, 7156) LDPC Code**

Each  $A_{i,j}$  is a  $511 \times 511$  circulant. The row weight of each of the 32 circulants is two; i.e., there are two '1's in each row. The total row weight of each row in the parity check matrix is  $2 \times 16$ , or 32. The column weight of each circulant is also two; i.e., there are two '1's in each column. The total weight of each column in the parity check matrix is  $2 \times 2$  or four. The position of the '1's in each circulant is defined in table 2-1. A scatter chart of the parity check matrix is shown in figure 2-4 where every '1' bit in the matrix is represented by a point.



**Figure 2-4: Scatter Chart of Parity Check Matrix**

**Table 2-1: Specification of Circulants**

| Circulant         | '1's position in 1 <sup>st</sup> row of circulant | Absolute '1's position in 1 <sup>st</sup> row of Parity Check Matrix |
|-------------------|---------------------------------------------------|----------------------------------------------------------------------|
| A <sub>1,1</sub>  | 0, 176                                            | 0, 176                                                               |
| A <sub>1,2</sub>  | 12, 239                                           | 523, 750                                                             |
| A <sub>1,3</sub>  | 0, 352                                            | 1022, 1374                                                           |
| A <sub>1,4</sub>  | 24, 431                                           | 1557, 1964                                                           |
| A <sub>1,5</sub>  | 0, 392                                            | 2044, 2436                                                           |
| A <sub>1,6</sub>  | 151, 409                                          | 2706, 2964                                                           |
| A <sub>1,7</sub>  | 0, 351                                            | 3066, 3417                                                           |
| A <sub>1,8</sub>  | 9, 359                                            | 3586, 3936                                                           |
| A <sub>1,9</sub>  | 0, 307                                            | 4088, 4395                                                           |
| A <sub>1,10</sub> | 53, 329                                           | 4652, 4928                                                           |
| A <sub>1,11</sub> | 0, 207                                            | 5110, 5317                                                           |
| A <sub>1,12</sub> | 18, 281                                           | 5639, 5902                                                           |
| A <sub>1,13</sub> | 0, 399                                            | 6132, 6531                                                           |
| A <sub>1,14</sub> | 202, 457                                          | 6845, 7100                                                           |
| A <sub>1,15</sub> | 0, 247                                            | 7154, 7401                                                           |
| A <sub>1,16</sub> | 36, 261                                           | 7701, 7926                                                           |
| A <sub>2,1</sub>  | 99, 471                                           | 99, 471                                                              |
| A <sub>2,2</sub>  | 130, 473                                          | 641, 984                                                             |
| A <sub>2,3</sub>  | 198, 435                                          | 1220, 1457                                                           |
| A <sub>2,4</sub>  | 260, 478                                          | 1793, 2011                                                           |
| A <sub>2,5</sub>  | 215, 420                                          | 2259, 2464                                                           |
| A <sub>2,6</sub>  | 282, 481                                          | 2837, 3036                                                           |
| A <sub>2,7</sub>  | 48, 396                                           | 3114, 3462                                                           |
| A <sub>2,8</sub>  | 193, 445                                          | 3770, 4022                                                           |
| A <sub>2,9</sub>  | 273, 430                                          | 4361, 4518                                                           |
| A <sub>2,10</sub> | 302, 451                                          | 4901, 5050                                                           |
| A <sub>2,11</sub> | 96, 379                                           | 5206, 5489                                                           |
| A <sub>2,12</sub> | 191, 386                                          | 5812, 6007                                                           |
| A <sub>2,13</sub> | 244, 467                                          | 6376, 6599                                                           |
| A <sub>2,14</sub> | 364, 470                                          | 7007, 7113                                                           |
| A <sub>2,15</sub> | 51, 382                                           | 7205, 7536                                                           |
| A <sub>2,16</sub> | 192, 414                                          | 7857, 8079                                                           |

Note that the numbers in the second column represent the relative column position of the '1's in the first row of each circulant. Since there are only 511 possible positions, these numbers can range only from 0 to 510. The third column represents the absolute position of the '1's in the parity-check matrix. There are exactly 8176 possible; therefore these numbers can range only from 0 to 8175.

### 2.3 ENCODING

The encoder can be designed using the method given in reference [C1.6]. The generator matrix of the (8176, 7156) code consists of two parts. The first part is a  $7154 \times 8176$  submatrix in systematic-circulant form as shown in figure 2-5. It consists of a  $7154 \times 7154$  identity matrix and two columns of  $511 \times 511$  circulants  $B_{i,j}$ s, each column consisting of 14 circulants. The Is are the  $511 \times 511$  identity submatrices and the 0s are the all zero  $511 \times 511$  submatrices. The second part consists of two independent rows. The first part generates a (8176, 7154) LDPC subcode of the (8176, 7156) code. Each codeword in the subcode consists of 7154 information bits and 1022 parity-check bits. For reason given in section 2.4, there are advantages in using the subcode implementation. The circulants  $B_{i,j}$ s are constructed based on the algorithm given below:

- 1) From figure 2-3 and table 2-1, define  $D = \begin{bmatrix} A_{1,15} & A_{1,16} \\ A_{2,15} & A_{2,16} \end{bmatrix}$ , which is a  $1022 \times 1022$  matrix.
- 2) Let  $u = (1 \ 0 \ 0 \ 0 \ \dots \ 0)$  be the unit 511 tuple, i.e., a vector quantity of length 511 with a '1' at the leftmost position and '0's in the rest.
- 3) Define  $z_i = (b_{i,1} \ b_{i,2})$ , where  $i = 1, 2, \dots, 14$  and the  $b_{i,j}$ s are the first row of the  $B_{i,j}$  circulants.
- 4) Define  $M_i = \begin{bmatrix} A_{1,i} \\ A_{2,i} \end{bmatrix}$ , where  $i = 1, 2, \dots, 14$ . (Note that the parity check matrix can now be represented as:  $[M_1 \ M_2 \ \dots \ M_{14} \ D]$ .)
- 5) Since the rank of  $D$  is 1020 and not 1022, there are two linearly dependent columns, 511<sup>th</sup> and 1022<sup>nd</sup>. Set the 511<sup>th</sup> and 1022<sup>nd</sup> elements of  $z_i$  to zero and solve  $M_i u^T + D z_i^T = 0$  for  $z_i$ , where  $i = 1, 2, \dots, 14$  and T superscript represents matrix transpose.
- 6) The  $b_{i,j}$ s can be extracted from the  $z_i$ s. (They are numerically tabulated in annex A1.)

There are many ways to design the encoder based on the generator matrix in figure 2-5. These schemes have complexities that are proportional to the length of the codeword or parity check bits (see reference [C1.6]).

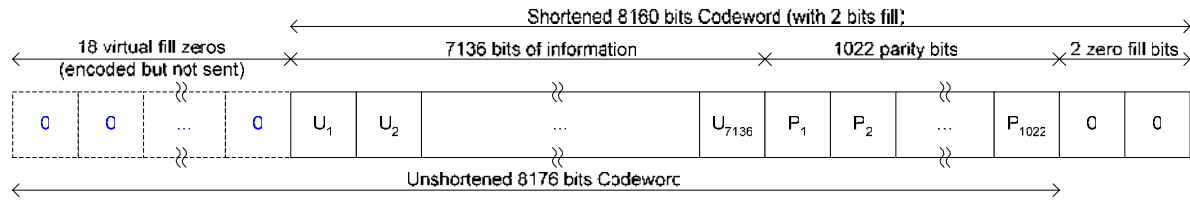
|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                   |                   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|-------------------|-------------------|
| I | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | B <sub>1,1</sub>  | B <sub>1,2</sub>  |
| 0 | I | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | B <sub>2,1</sub>  | B <sub>2,2</sub>  |
| 0 | 0 | I | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | B <sub>3,1</sub>  | B <sub>3,2</sub>  |
| 0 | 0 | 0 | I | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | B <sub>4,1</sub>  | B <sub>4,2</sub>  |
| 0 | 0 | 0 | 0 | I | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | B <sub>5,1</sub>  | B <sub>5,2</sub>  |
| 0 | 0 | 0 | 0 | 0 | I | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | B <sub>6,1</sub>  | B <sub>6,2</sub>  |
| 0 | 0 | 0 | 0 | 0 | 0 | I | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | B <sub>7,1</sub>  | B <sub>7,2</sub>  |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | I | 0 | 0 | 0 | 0 | 0 | 0 | 0 | B <sub>8,1</sub>  | B <sub>8,2</sub>  |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | I | 0 | 0 | 0 | 0 | 0 | 0 | B <sub>9,1</sub>  | B <sub>9,2</sub>  |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | I | 0 | 0 | 0 | 0 | 0 | B <sub>10,1</sub> | B <sub>10,2</sub> |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | I | 0 | 0 | 0 | 0 | B <sub>11,1</sub> | B <sub>11,2</sub> |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | I | 0 | 0 | 0 | B <sub>12,1</sub> | B <sub>12,2</sub> |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | I | 0 | 0 | B <sub>13,1</sub> | B <sub>13,2</sub> |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | I | 0 | B <sub>14,1</sub> | B <sub>14,2</sub> |

**Figure 2-5: Systematic Circulant Generator Matrix**

## 2.4 SHORTENED (8160, 7136) CODE

Using the generator matrix given by figure 2-5, an encoder can be implemented using a circuit described in reference [C1.6]. This encoder generates a (8176, 7154) LDPC subcode of the (8176, 7156) code. Current spacecraft and ground systems manipulate and process data at 32-bit computer word size. Neither (8176, 7154) nor (8176, 7156) is a multiple of 32. It is beneficial to shorten the codeword to the dimensions of (8160, 7136). In other words, by shortening the information sequence to 7136 through the use of 18 bits of virtual fill, the (8176, 7154) subcode encoder can be used. This is accomplished by encoding the virtual fill bits with zeros but not transmitting them; thus the total codeword length becomes 8158. Note that it is not necessary to add two independent rows to the generator matrix to encode the full (8176, 7156) code because these bits would be shortened anyway, and so the subcode is sufficient and less complicated for this application. Since the code length of 8158 is two bits shy of 8160, an exact multiple of 32, two bits of actual transmitted zero fill are appended to the end of the codeword to achieve a shortened code dimension of (8160, 7136) bits, or (1020, 892) octets, or (255, 223) 32-bit words. The shortened codeword is shown in figure 2-6.

The received shortened codeword would require the removal of the two zero fill bits prior to decoding. The decoder would then reproduce the 18 virtual fill zeros after processing but would, in general, not pass these 18 zeros on to the ground equipment.



**Figure 2-6: Shortened Codeword**

## 2.5 RANDOMIZATION AND SYNCHRONIZATION

The use of the shortened (8160, 7136) LDPC code does not guarantee sufficient bit (symbol) transitions to acquire or maintain bit (symbol) synchronization. It is highly recommended that a pseudo-randomizer be used after encoding in accordance with section 7 of reference [1].

In addition, frame (codeword) synchronization is required so that the receiver can identify the beginning of the frame (codeword) for proper decoding. The use of an attached sync marker (ASM) as specified in section 6.6 of reference [1] is required. Note that the ASM is not pseudo-randomized.

### 3 LOW DENSITY PARITY CHECK CODE FAMILY OPTIMIZED FOR DEEP SPACE APPLICATIONS

#### 3.1 OVERVIEW

##### 3.1.1 BACKGROUND

The Low-Density Parity-Check (LDPC) codes presented in this document are intended to complement the current codes in the CCSDS Recommended Standard, *TM Synchronization and Channel Coding* (reference [1]), and were designed according to a list of requirements and evaluation criteria that reflect the needs of spacecraft applications (see reference [C2.2]).

- Requirements
  - *Code rates*: The family shall include codes of rate  $\approx 0.5$  and  $\approx 0.8$
  - *Block lengths*: The family shall cover  $k \approx 1000$  to  $k \approx 16000$  information bits spaced by multiples of  $\approx 4$
  - *Family*: A single hardware decoder shall be appropriate for all codes
  - *Intellectual property*: There must be no restrictions for CCSDS members
- Desired Properties
  - *Systematic encoders*: Systematic encoders are preferred
  - *Code rates*: One or two intermediate rates from  $\{1/2, 2/3, 3/4, 4/5\}$  are desired
- Evaluation Criteria
  - *Decoder computation*: Codes requiring fewer decoder message computations are preferred
  - *Encoder computation*: Preferred encoders require fewer logic gates for a given speed
  - *Descriptive complexity*: The code description in a standards document should be short
  - *Code performance*: Codes requiring less  $E_b/N_0$  at Word Error Rate (WER) =  $10^{-4}$  and  $10^{-6}$  are preferred

The selected code rates are 1/2, 2/3, and 4/5, three values, which are about uniformly spaced by 1 dB on the rate-dependent capacity curve for the binary-input Additive White Gaussian Noise (AWGN) channel (reference [C2.4]). Near rate 1/2, a 1% improvement in bandwidth efficiency costs about 0.02 dB in power efficiency; near rate 7/8, a 1% improvement in bandwidth efficiency costs 0.1 dB in power efficiency. Hence, the use of a higher order modulation may be a more practical means for saving bandwidth than the use of a code with

rate much above 0.8. The code rates are exact ratios of small integers to simplify implementation.

The selected block lengths are  $k=1024$ ,  $k=4096$ , and  $k=16384$ . The three values  $k=\{1024,4096,\infty\}$  are about uniformly spaced by 0.6 dB on the sphere-packing bound at  $W_{ER}=10^{-8}$ , and reducing the last value from  $\infty$  to 16384 makes the largest block size practical at a cost of about 0.3 dB. By choosing to keep  $k$  constant among family members, rather than  $n$ , the spacecraft's command and data handling system can generate data frames without knowledge of the code rate. Choosing powers of 2 may simplify implementation.

Implementers should be aware that many patents have been filed on LDPC codes; in particular, a procedure for parallelized decoding of LDPC codes is covered by a U.S. patent granted to T. Richardson and V. Novichkov.<sup>1</sup>

The selected codes are systematic. Two low-complexity encoding methods are described (see reference [C2.3]), and either can be used, depending on constraints of the chosen technology (software, FPGA, ASIC). The parity check matrices have plenty of structure to facilitate decoder implementation (see reference [C2.6]). The codes have irregular degree distributions, because this improves performance by about 0.5 dB at rate 1/2, compared to a regular (3,6) code (see reference [C2.1]).

### 3.1.2 DESCRIPTION OF THE CODE

Like turbo codes, low-density parity-check (LDPC) codes are binary block codes with large code blocks (hundreds or thousands of bits). They may be systematic or non-systematic, and they may be transparent or non-transparent. The nine LDPC codes defined here are systematic. All are non-transparent,<sup>2</sup> and phase ambiguities are resolved using frame markers, which are required for Codeblock synchronization.

Like turbo codes, LDPC codes may be used to obtain greater coding gain than those provided by concatenated coding systems. In contrast to turbo codes, LDPC codes offer the prospect of much higher decoding speeds via highly parallelized decoder structures. Currently reference [1] includes turbo codes of rates 1/2 and lower and a convolutional code rate 1/2. LDPC codes of rates 1/2 and higher are defined in this experimental specification. Therefore rate 1/2 is the only rate at which this specification and reference [1] are comparable.

---

<sup>1</sup> Richardson, Tom, and Novichkov, Vladimir. 2001. Methods and apparatus for decoding LDPC codes. U.S. Patent 6,633,856, filed October 10, 2001, and issued October 14, 2003.

<sup>2</sup>Differential encoding (i.e., NRZ-M signaling) after the LDPC encoder is not recommended since soft decoding would require the use of differential detection with considerable loss of performance. Differential encoding before the LDPC encoder cannot be used because the LDPC codes specified in this document are non-transparent. This implies that phase ambiguities have to be detected and resolved by the frame synchronizer.

## NOTES

- 1 LDPC coding, by itself, cannot guarantee sufficient bit transitions to keep receiver symbol synchronizers in lock. Therefore, the Pseudo-Randomizer defined in section 3.5 is required unless the system designer verifies that sufficient symbol transition density is assured by other means when the Randomizer is not used.
- 2 While providing outstanding coding gain, LDPC codes generally may still leave some residual errors in the decoded output. For this reason, when CCSDS TM or AOS Transfer Frames are used, references [C2.7] and [C2.8], respectively, require that a Cyclic Redundancy Check (CRC) be used to validate the frame.

### 3.2 SPECIFICATION

An LDPC code is specified indirectly by a  $v$ -by- $w$  parity-check matrix  $H$  consisting of  $v$  linearly independent rows. A coded sequence of  $w$  bits must satisfy all  $v$  parity-check equations corresponding to the  $v$  rows of  $H$ . Parity-check matrices may include additional linearly dependent rows without changing the code. An encoder maps an input frame of  $k \leq w-v$  information bits uniquely into a codeblock of  $n \leq w$  bits. If  $n < w$ , the remaining  $w-n$  code symbols are punctured and are not transmitted. If  $k < w-v$ , the remaining dimensions of the code remain unused.

The codeblock lengths  $n$  and information block lengths  $k$ , and the corresponding rates  $r=k/n$ , are shown in table 3-1 for the suite of LDPC codes. The LDPC code rates  $r$  are exactly as indicated in table 3-1, unlike the case of turbo codes for which the precise code rates are slightly lower than the corresponding nominal rates due to termination bits.

**Table 3-1: Codeblock Lengths for Supported Code Rates (Measured in Bits)**

| Information block length $k$ | Code block length $n$ |          |          |
|------------------------------|-----------------------|----------|----------|
|                              | rate 1/2              | rate 2/3 | rate 4/5 |
| 1024                         | 2048                  | 1536     | 1280     |
| 4096                         | 8192                  | 6144     | 5120     |
| 16384                        | 32768                 | 24576    | 20480    |

**Table 3-2: Values of Submatrix Size  $M$  for Supported Codes**

| Information block length $k$ | Submatrix size $M$ |          |          |
|------------------------------|--------------------|----------|----------|
|                              | rate 1/2           | rate 2/3 | rate 4/5 |
| 1024                         | 512                | 256      | 128      |
| 4096                         | 2048               | 1024     | 512      |
| 16384                        | 8192               | 4096     | 2048     |

For each  $(n,k)$  in table 3-1, both the parity-check matrix  $H$  and the encoding rule for generating  $n$  code bits from  $k$  information bits are specified.

### 3.3 PARITY CHECK MATRICES

The  $H$  matrices are constructed from  $M \times M$  submatrices, where the submatrix size is listed in table 3-2.

The  $H$  matrices for the rate-1/2 codes are specified as follows,

$$H_{1/2} = \begin{bmatrix} 0_M & 0_M & 0_M & I_M & I_M \oplus \Pi_1 \\ I_M \oplus \Pi_8 & I_M \oplus \Pi_7 \oplus \Pi_6 & 0_M & 0_M & I_M \\ 0_M & I_M & I_M \oplus \Pi_5 & 0_M & \Pi_4 \oplus \Pi_3 \oplus \Pi_2 \end{bmatrix}$$

where  $I_M$  and  $0_M$  are the  $M \times M$  identity and zero matrices, respectively, and  $\Pi_1$  through  $\Pi_8$  are permutation matrices. The  $H$  matrices for the rate-2/3 and rate-4/5 codes are specified with additional columns and permutation matrices as follows.

$$H_{2/3} = \left[ \begin{array}{cc|c} 0_M & 0_M & \\ I_M & \Pi_{11} \oplus \Pi_{10} \oplus \Pi_9 & \\ \hline \Pi_{14} \oplus \Pi_{13} \oplus \Pi_{12} & I_M & H_{1/2} \end{array} \right]$$

$$H_{4/5} = \left[ \begin{array}{cccc|c} 0_M & 0_M & 0_M & 0_M & \\ I_M & \Pi_{23} \oplus \Pi_{22} \oplus \Pi_{21} & I_M & \Pi_{17} \oplus \Pi_{16} \oplus \Pi_{15} & \\ \hline \Pi_{26} \oplus \Pi_{25} \oplus \Pi_{24} & I_M & \Pi_{20} \oplus \Pi_{19} \oplus \Pi_{18} & I_M & H_{2/3} \end{array} \right]$$

Permutation matrix  $\Pi_k$  has non-zero entries in row  $i$  and column  $\pi_k(i)$  for  $i \in \{0, \dots, M-1\}$  and

$$\pi_k(i) = \frac{M}{4} \left( (\theta_k + \lfloor 4i/M \rfloor) \bmod 4 \right) + \left( \phi_k(\lfloor 4i/M \rfloor) + i \right) \bmod \frac{M}{4}$$

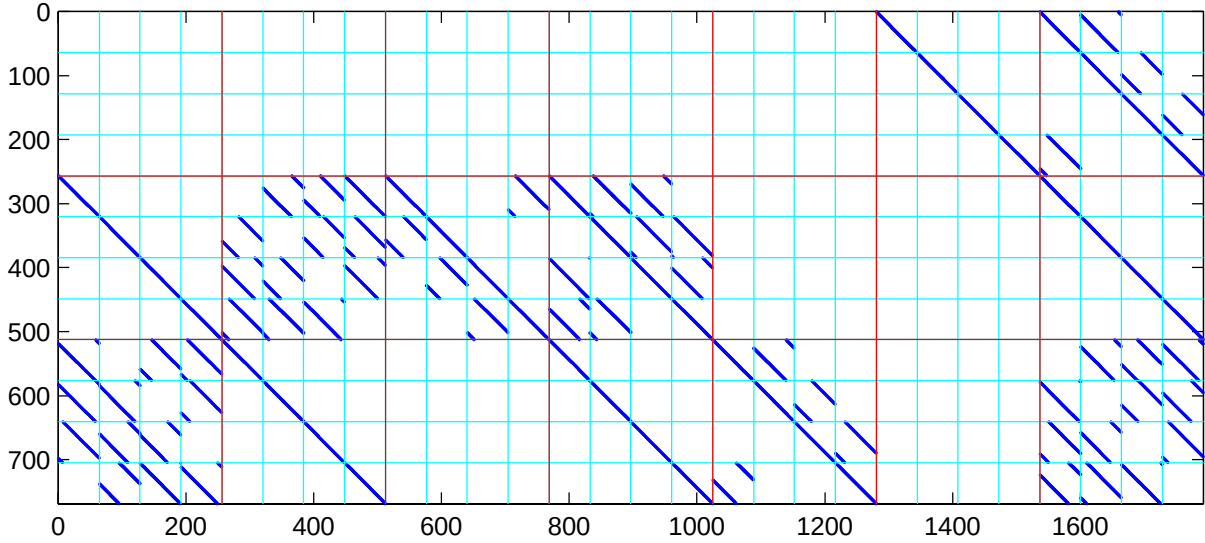
where the functions  $\theta_k$  and  $\phi_k(j)$  are defined by:

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|     | $\theta_k$ | $\phi_k(0)$ | $\phi_k(1)$ | $\phi_k(2)$ | $\phi_k(3)$ |
|-----|------------|-------------|-------------|-------------|-------------|
| k=1 | 1          | 1787        | 1502        | 1887        | 1291        |
| 2   | 1          | 1077        | 602         | 521         | 301         |
| 3   | 2          | 1753        | 749         | 590         | 1353        |
| 4   | 3          | 697         | 1662        | 1775        | 1405        |
| 5   | 1          | 1523        | 1371        | 1738        | 997         |
| 6   | 1          | 5           | 9           | 2032        | 2032        |
| 7   | 2          | 2035        | 131         | 2047        | 11          |
| 8   | 3          | 331         | 1884        | 85          | 1995        |
| 9   | 1          | 1920        | 1268        | 1572        | 623         |
| 10  | 2          | 130         | 1784        | 78          | 73          |
| 11  | 3          | 4           | 19          | 26          | 1839        |
| 12  | 0          | 85          | 1839        | 298         | 2003        |
| 13  | 2          | 551         | 81          | 1177        | 2019        |
| 14  | 3          | 15          | 2031        | 1950        | 1841        |
| 15  | 0          | 1780        | 76          | 1806        | 167         |
| 16  | 2          | 1960        | 336         | 128         | 1087        |
| 17  | 3          | 3           | 529         | 1855        | 2032        |
| 18  | 0          | 145         | 74          | 129         | 388         |
| 19  | 1          | 1019        | 68          | 269         | 1385        |
| 20  | 3          | 691         | 186         | 1614        | 885         |
| 21  | 0          | 132         | 905         | 1467        | 707         |
| 22  | 1          | 42          | 1751        | 1533        | 1272        |
| 23  | 3          | 393         | 1516        | 925         | 7           |
| 24  | 0          | 502         | 1285        | 1886        | 1534        |
| 25  | 1          | 201         | 1597        | 2046        | 1965        |
| 26  | 2          | 1064        | 1712        | 1167        | 588         |

These parity check matrices have one redundant check equation: the last  $2M$  rows sum to zero (modulo 2). The last  $M$  code symbols are punctured. One dimension of the code is not used.

For example, the parity check matrix for the ( $n=1536$ ,  $k=1024$ ) code is shown in figure 3-1 with dots representing each of the non-zero entries, and its structure is indicated by gridlines.



**Figure 3-1: An  $H$  Matrix for the  $(n=1536, k=1024)$  Code**

### 3.4 ENCODERS

The encoding rule for producing codeblocks consistent with these  $H$  matrices is as follows. The input message is  $(m_0, m_1, \dots, m_{k-1})$  and the symbol  $\oplus$  denotes the exclusive OR operation. The submatrix size is  $M$  as listed in table 3-2.

The five encoding steps are:

- 1) For  $0 \leq j \leq 7$ , define the vectors:

$$s^j = \begin{cases} (m_{jM}, \dots, m_{(j+1)M-1}) & \text{if } j < k/M \\ (0, \dots, 0), \text{ the length } - M \text{ zero vector} & \text{otherwise} \end{cases}$$

Thus, potentially non-zero vectors are  $s^0$  and  $s^1$  for rate 1/2 codes,  $s^0$  through  $s^3$  for rate 2/3 codes, and  $s^0$  through  $s^7$  for rate 4/5 codes.

- 2) Compute the vector  $p^2$  of length  $M$  by sparse matrix multiplication:

$$p_i^2 = s_i^0 \oplus s_{\pi_6(i)}^0 \oplus s_{\pi_7(i)}^0 \oplus s_i^1 \oplus s_{\pi_8(i)}^1 \oplus s_{\pi_9(i)}^2 \oplus s_{\pi_{10}(i)}^2 \oplus s_{\pi_{11}(i)}^3 \oplus s_i^3 \\ \oplus s_{\pi_{15}(i)}^4 \oplus s_{\pi_{16}(i)}^4 \oplus s_{\pi_{17}(i)}^4 \oplus s_i^5 \oplus s_{\pi_{21}(i)}^6 \oplus s_{\pi_{22}(i)}^6 \oplus s_{\pi_{23}(i)}^6 \oplus s_i^7$$

- 3) Compute the non-accumulated portion of the parity  $p^1$  by sparse matrix multiplication:

$$p_i^1 = p_i^2 \oplus p_{\pi_1(i)}^2$$

- 4) Compute the accumulated portion of the parity,  $p^0$ , by initializing:

$$j_0 = 0 \quad \text{and} \quad p_0^0 = 0$$

For  $i \in \{0, \dots, M-1\}$ , compute,

$$j_{i+1} = \pi_5(j_i)$$

$$\begin{aligned} p_{j_{i+1}}^0 = & p_{j_i}^0 \oplus p_{\pi_2(j_i)}^2 \oplus p_{\pi_3(j_i)}^2 \oplus p_{\pi_4(j_i)}^2 \oplus s_{j_i}^0 \oplus s_{j_i}^2 \oplus s_{\pi_{12}(j_i)}^3 \oplus s_{\pi_{13}(j_i)}^3 \oplus s_{\pi_{14}(j_i)}^3 \\ & \oplus s_{j_i}^4 \oplus s_{\pi_{18}(j_i)}^5 \oplus s_{\pi_{19}(j_i)}^5 \oplus s_{\pi_{20}(j_i)}^5 \oplus s_{j_i}^6 \oplus s_{\pi_{24}(j_i)}^7 \oplus s_{\pi_{25}(j_i)}^7 \oplus s_{\pi_{26}(j_i)}^7 \end{aligned}$$

- 5) Form the output codeword  $c$ :

$$c = \begin{cases} (s^1, s^0, p^0, p^1) & \text{if } r = 1/2 \\ (s^3, s^2, s^1, s^0, p^0, p^1) & \text{if } r = 2/3 \\ (s^7, s^6, s^5, s^4, s^3, s^2, s^1, s^0, p^0, p^1) & \text{if } r = 4/5 \end{cases}$$

### 3.5 SYNCHRONIZATION

Codeblock synchronization is achieved by synchronization of an Attached Sync Marker associated with each LDPC Codeblock. The Attached Sync Marker (ASM) is a bit pattern specified in section 6 of CCSDS Recommended Standard CCSDS 131.0-B-1, *TM Synchronization and Channel Coding* (reference [1]), as an aid to synchronization, and it precedes the LDPC Codeblock. Frame synchronizers should be set to expect a marker at a recurrence interval equal to the length of the ASM plus that of the Turbo Codeblock. All codes in the LDPC family use the 64-bit ASM.

## ANNEX A

### ANNEX TO SECTION 2, LOW DENSITY PARITY CHECK CODE OPTIMIZED FOR NEAR-EARTH APPLICATIONS

#### A1 GENERATOR MATRIX CIRCULANT TABLE

**Table A-1: Table of Circulants for the Generator Matrix**

| Circulant        | 1 <sup>st</sup> row of circulant                                                                                                 |
|------------------|----------------------------------------------------------------------------------------------------------------------------------|
| B <sub>1,1</sub> | 55BF56CC55283DFEEFEA8C8CFF04E1EBD9067710988E25048D67525426939E2068D2DC6FCD2F822BEB6BD96C8A76F4932AAE9BC53AD20A2A9C86BB461E43759C |
| B <sub>1,2</sub> | 6855AE08698A50AA3051768793DC238544AF3FE987391021AAF6383A6503409C3CE971A80B3ECE12363EE809A01D91204F1811123EAB867D3E40E8C652585D28 |
| B <sub>2,1</sub> | 62B21CF0AEE0649FA67B7D0EA6551C1CD194CA77501E0FCF8C85867B9CF679C18BCF7939E10F8550661848A4E0A9E9EDB7DAB9EDABA18C168C8E28AACDDEAB1E |
| B <sub>2,2</sub> | 64B71F486AD57125660C4512247B229F0017BA649C6C11148FB00B70808286F1A9790748D296A593FA4FD2C6D7AAF7750F0C71B31AEE5B400C7F5D73AAF00710 |
| B <sub>3,1</sub> | 681A8E51420BD829ACE13E491D618083FFBBA830DB5FAF330209877D801F92B5E07117C57E75F6F0D873B3E520F21EAFD78C1612C6228111A369D5790F5929A  |
| B <sub>3,2</sub> | 04DF1DD77F1C20C1FB570D7DD7A1219EAECEA4B2877282651B0FFE713DF338A63263BC0E324A87E2DC1AD64C9F10AAA585ED6905946EE167A73CF04AD2AF9218 |
| B <sub>4,1</sub> | 35951FEE6F20C902296C9488003345E6C5526C5519230454C556B8A04FC0DC642D682D94B4594B5197037DF15B5817B26F16D0A3302C09383412822F6D2B234E |
| B <sub>4,2</sub> | 7681CF7F278380E28F1262B22F40BF3405BFB92311A8A34D084C086464777431DBFDDD2E82A2E6742BAD6533B51B2BDEE0377E9F6E63DCA0B0F1DF97E73D5CD8 |
| B <sub>5,1</sub> | 188157AE41830744BAE0ADA6295E08B79A44081E111F69BBE7831D07BEEBF76232E065F752D4F218D39B6C5BF20AE5B8FF172A7F1F680E6BF5AAC3C4343736C2 |
| B <sub>5,2</sub> | 5D80A6007C175B5C0DD88A442440E2C29C6A136BBCE0D95A58A83B48CA0E7474E9476C92E33D164BFF943A61CE1031DFF441B0B175209B498394F4794644392E |
| B <sub>6,1</sub> | 60CD1F1C282A1612657E8C7C1420332CA245C0756F78744C807966C3E1326438878BD2CCC83388415A612705AB192B3512EEF0D95248F7B73E5B0F412BF76DB4 |
| B <sub>6,2</sub> | 434B697B98C9F3E48502C8DBD891D0A0386996146DEBEF11D4B833033E05EDC28F808F25E8F314135E6675B7608B66F7FF3392308242930025DDC4BB65CD7B6E |
| B <sub>7,1</sub> | 766855125CFDC804DAF8DBE3660E8686420230ED4E049DF11D82E357C54FE256EA01F5681D95544C7A1E32B7C30A8E6CF5D0869E754FFDE6AEFA6D7BE8F1B148 |
| B <sub>7,2</sub> | 222975D325A487FE560A6D146311578D9C5501D28BC0A1FB48C9BDA173E869133A3AA9506C42AE9F466E85611FC5F8F74E439638D66D2F00C682987A96D8887C |
| B <sub>8,1</sub> | 14B5F98E8D55FC8E9B4EE453C6963E052147A857AC1E08675D99A308E7269FAC5600D7B155DE8CB1BAC786F45B46B523073692DE745FDF10724DDA38FD09B1C  |

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| <b>Circulant</b>  | <b>1<sup>st</sup> row of circulant</b>                                                                                                |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| B <sub>8,2</sub>  | 1B71AFFB8117BCF8B5D002A99FEEA49503C0359B056963FE5271140E626F6F8FC<br>E9F29B37047F9CA89EBCE760405C6277F329065DF21AB3B779AB3E8C8955400  |
| B <sub>9,1</sub>  | 0008B4E899E5F7E692BDCE69CE3FAD997183CFAEB2785D0C3D9CAE510316D4BD<br>65A2A06CBA7F4E4C4A80839ACA81012343648EEA8DBBA2464A68E115AB3F4034  |
| B <sub>9,2</sub>  | 5B7FE6808A10EA42FEF0ED9B41920F82023085C106FBBBC1F56B567A14257021BC5<br>FDA60CBA05B08FAD6DC3B0410295884C7CCDE0E56347D649DE6DDCEEB0C95E |
| B <sub>10,1</sub> | 5E9B2B33EF82D0E64AA2226D6A0ADCD179D5932EE1CF401B336449D0FF775754C<br>A56650716E61A43F963D59865C7F017F53830514306649822CAA72C152F6EB2  |
| B <sub>10,2</sub> | 2CD8140C8A37DE0D0261259F63AA2A420A8F81FECB661DBA5C62DF6C817B4A61<br>D2BC1F068A50DFD0EA8FE1BD387601062E2276A4987A19A70B460C54F215E184  |
| B <sub>11,1</sub> | 06F1FF249192F2EAF063488E267EEE994E7760995C4FA6FFA0E4241825A7F5B65C<br>74FB16AC4C891BC008D33AD4FF97523EE5BD14126916E0502FF2F8E4A07FC2  |
| B <sub>11,2</sub> | 65287840D00243278F41CE1156D1868F24E02F91D3A1886ACE906CE741662B40B4<br>EFD9B90F76C1ADD884D920AFA8B3427EEB84A759FA02E00635743F50B942F0  |
| B <sub>12,1</sub> | 4109DA2A24E41B1F375645229981D4B7E88C36A12DAB64E91C764CC43CCEC188E<br>C8C5855C8FF488BB91003602BEF43DBEC4A621048906A2CDC5DBD4103431DB8  |
| B <sub>12,2</sub> | 2185E3BC7076BA51AAD6B199C8C60BCD70E8245B874927136E6D8DD527DF0693<br>DC10A1C8E51B5BE93FF7538FA138B335738F4315361ABF8C73BF40593AE22BE4  |
| B <sub>13,1</sub> | 228845775A262505B47288E065B23B4A6D78AFBDDDB2356B392C692EF56A35AB4A<br>A27767DE72F058C6484457C95A8CCDD0EF225ABA56B7657B7F0E947DC17F972 |
| B <sub>13,2</sub> | 2630C6F79878E50CF5ABD353A6ED80BEACC7169179EA57435E44411BC7D566136<br>DFA983019F3443DE8E4C60940BC4E31DCEAD514D755AF95A622585D69572692  |
| B <sub>14,1</sub> | 7273E8342918E097B1C1F5FEF32A150AEF5E11184782B5BD5A1D8071E94578B0AC<br>722D7BF49E8C78D391294371FFBA7B88FABF8CC03A62B940CE60D669DFB7B6  |
| B <sub>14,2</sub> | 087EA12042793307045B283D7305E93D8F74725034E77D25D3FF043ADC5F8B5B18<br>6DB70A968A816835EFB575952EAE7EA4E76DF0D5F097590E1A2A978025573E  |

Note that the numbers in the second column represent the hexadecimal representation of the first row of each circulant. Since there are only 511 possible positions, the leftmost bit is padded with a zero to allow a 128 digit hexadecimal number. Table A-1 cannot be as efficiently described as table 2-5 because the generator circulants do not have a low density of '1's.

## A2 COMPLEXITY

The complexity of LDPC codes has been an area of research and discussion. For a Field Programmable Gate Array (FPGA) or Application Specific Integrated Circuit (ASIC) implementation, the encoder's complexity are dominated by two factors: 1) the total number of required logic gates and 2) the routing complexity. For the code presented in this Experimental Specification, the quasi-cyclic property allows for the use of shift registers whose required number of logic gates is proportional to  $n-k$  (see reference [C1.6]) or  $8176-7156 = 1020$  (unshortened). With regard to the routing complexity, there is currently no way

to predict this figure: it would depend on a number of factors, such as the choice of the FPGA or ASIC, the routing algorithm, and the layout of the device.

The decoder's complexity is larger than the encoder's and even more difficult to predict. The primary complexity factors (the total number of required logic gates and the routing complexity) are a function of the choice of BP decoding algorithm (there are many) as well as the architectural decisions (i.e., parallel or serial processing, number of bits of finite precision, fixed number of iterations or stopping rule, use of look-up tables, etc.) These choices also determine the decoder's Bit Error Rate (BER) performance.

For the development of the baselined (8176, 7156) code, an FPGA implementation was used to confirm the software simulations. A Xilinx 8000 Virtex-2 FPGA was used for the test. The device contained both the encoder and decoder. The decoder algorithm was a Scaled Min-Sum Parallel BP Decoder (SMSPD) described in reference [C1.7]. The encoder algorithm was a shift register based encoder described in reference [C1.6]. An architectural evaluation was performed prior to implementation to produce a quasi-optimal implementation based on routing, logic requirements, and BER performance.

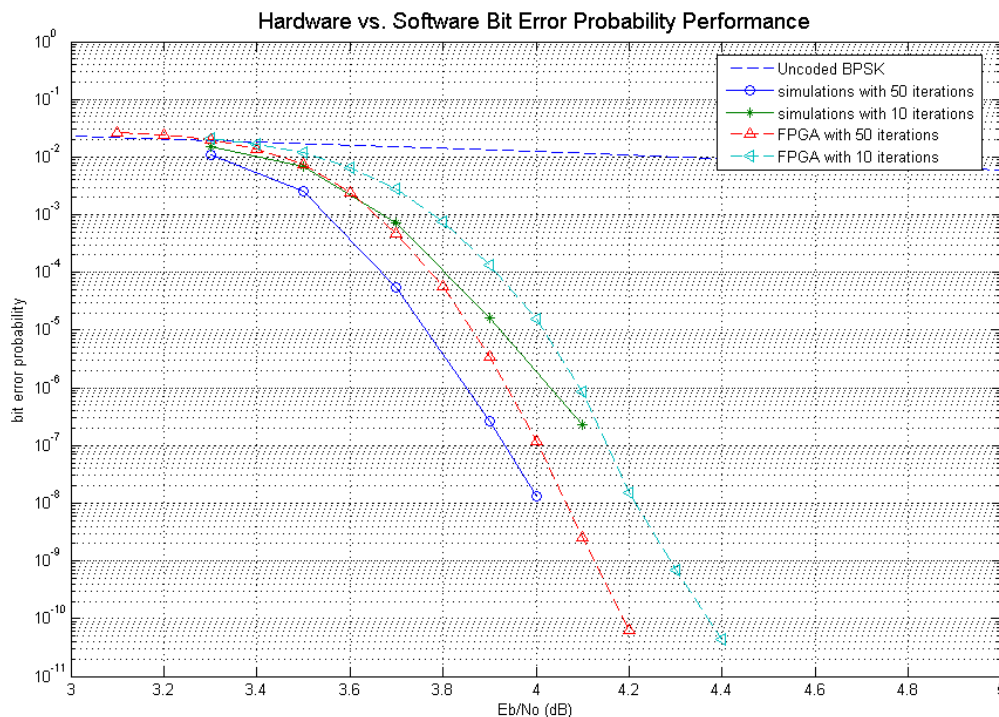
The FPGA had the following statistics:

- 1) encoder used 2,535 logic slices out of 46,592 available or 5.4% and 4 memory blocks out of 168 available or 2.4%;
- 2) decoder used 21,803 logic slices out of 46,592 or 46.8% and 137 memory blocks out of 168 or 81.5%.

The number of logic slices is an aggregate measure of the number of logic gates required and the routing complexity, while the memory blocks figure is the number of dedicated FPGA memory blocks used. It is clear from these statistics that the encoder is of much lower complexity than the decoder, using only 5.4% of the logic slices resources while the decoder requires 46%.

Annex A3 summarizes the test results.

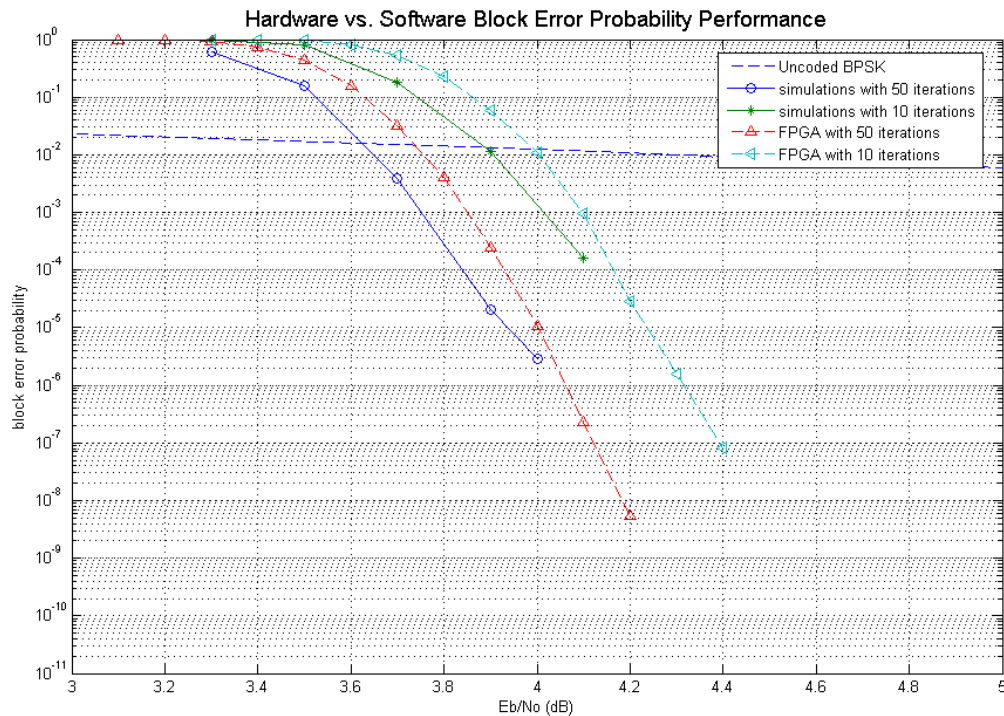
### A3 FPGA TEST RESULTS



**Figure A-1: Bit Error Rate Test Results**

Figure A-1 shows the BER and figure A-2 shows the Block Error Rate (BLER) test results for 50 and 10 maximum iterations from an FPGA implementation of the baselined (8176, 7156) code. Note that for both cases the difference between simulations and hardware tests was 0.1 dB or less.

The encoder data rate was limited to  $2 \times$  system clock while the decoder operated at  $14 \times$  system clock / number of iterations. For testing, the system clock was set to 100 MHz, so for 10 iterations, the decoder operated at 140 Mbps. Although the shortened (8160, 7136) was not tested, it is reasonable to say that the baselined (8176, 7156) and the shortened (8160, 7136) codes will have similar results.



**Figure A-2: Block Error Rate Test Results**

## ANNEX B

### ANNEX TO SECTION 3, LOW DENSITY PARITY CHECK CODE FAMILY OPTIMIZED FOR DEEP SPACE APPLICATIONS

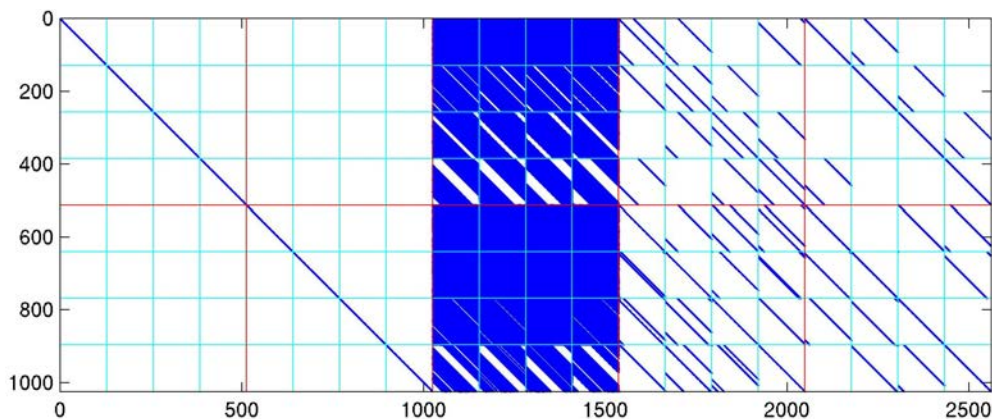
#### B1 QUASICYCLIC ENCODERS

The encoding algorithm described in section 3.4 can be appropriate for microprocessor-based encoders that operate on many bits in parallel. For a hardware bit-serial encoder, such as might be implemented in an FPGA or ASIC, matrix multiplication by a (dense) quasicyclic generator matrix  $G_{qc}$  may be more convenient.

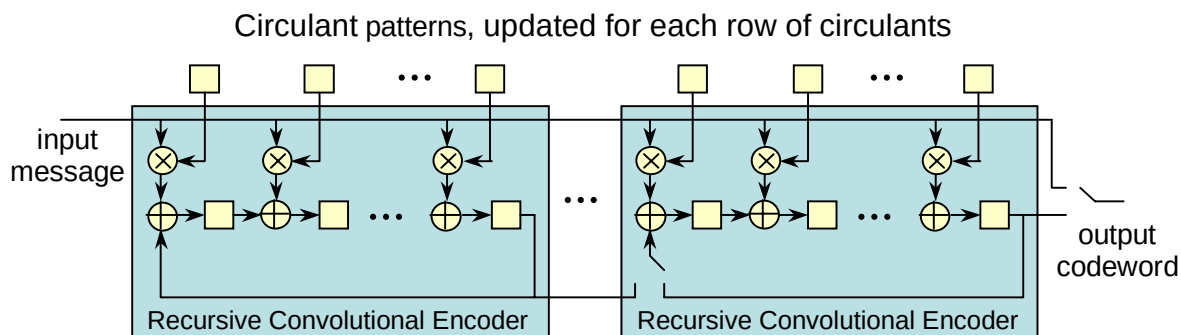
For the codes of rate 1/2, 2/3, and 4/5, quasicyclic generator matrices  $G_{qc}$  may be constructed as follows. Let  $K=2$  for a rate 1/2 code,  $K=4$  for rate 2/3, and  $K=8$  for rate 4/5, so  $G_{qc}$  will be of size  $MK \times M(K+2)$ .

- 1) Let  $P$  be the  $3M-1 \times 3M-1$  submatrix of  $H$  consisting of the first  $3M-1$  rows of the last  $3M-1$  columns. Let  $Q$  be the  $3M-1 \times MK$  submatrix of  $H$  consisting of the first  $3M-1$  rows of the first  $MK$  columns.
- 2) Compute  $W = (P^{-1}Q)^T$ , where the arithmetic is performed modulo-2.
- 3) Construct the matrix  $G = [I_{MK} \quad 0_{MK \times 1} \quad W]$ , where  $I_{MK}$  is the  $MK \times MK$  identity matrix, and  $0_{MK \times 1}$  is the  $MK \times 1$  all-zero column vector. This generator matrix defines the same encoder as that described in section 3.4 (when the last  $M$  columns are deleted, corresponding to the untransmitted vector  $p^2$ ), and it is not quasicyclic. Note that the all-zero column in  $G$  defines a code symbol that is always zero, as does the initialization  $p_0^1=0$  in Step 4) of section 3.4.
- 4) Let  $G_{qc}$  be the same size as  $G$ , with rows 1,  $1+M/4$ ,  $1+2M/4$ , ...,  $1+MK-M/4$  identical to the corresponding rows of  $G$ . These are the first rows of circulants of size  $M/4$ ; the remaining rows of  $G_{qc}$  complete the circulants. In implementation, the circulants in the last  $M$  columns are deleted, for these code symbols are punctured.

For example, the generator matrix  $G_{qc}$  for the  $(n=2048, k=1024)$  code is shown in figure B-1, including the last  $M$  columns for the punctured symbols.



**Figure B-1: A Quasicyclic Generator Matrix for the (2048,1024) Code**

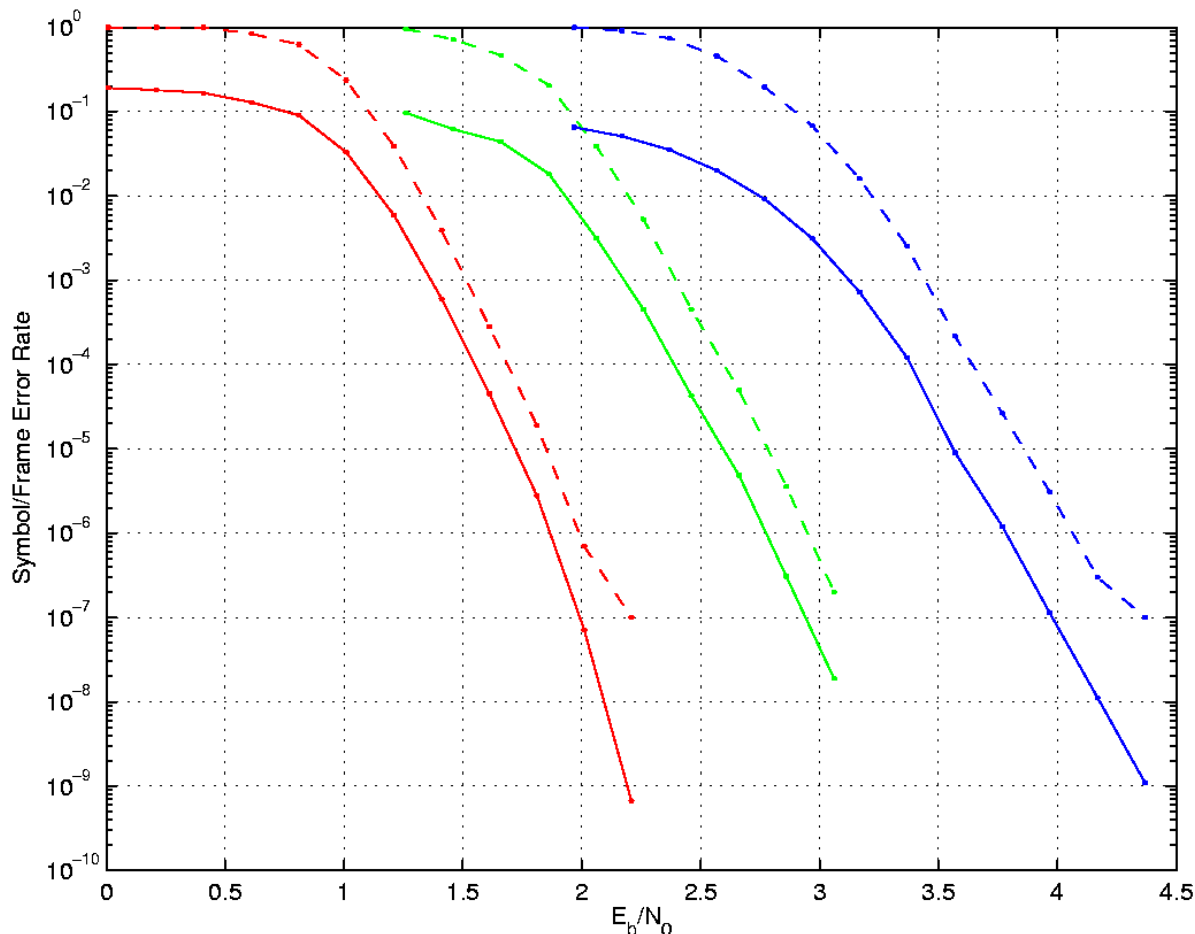


**Figure B-2: A Quasicyclic Encoder Using Feedback Shift Registers**

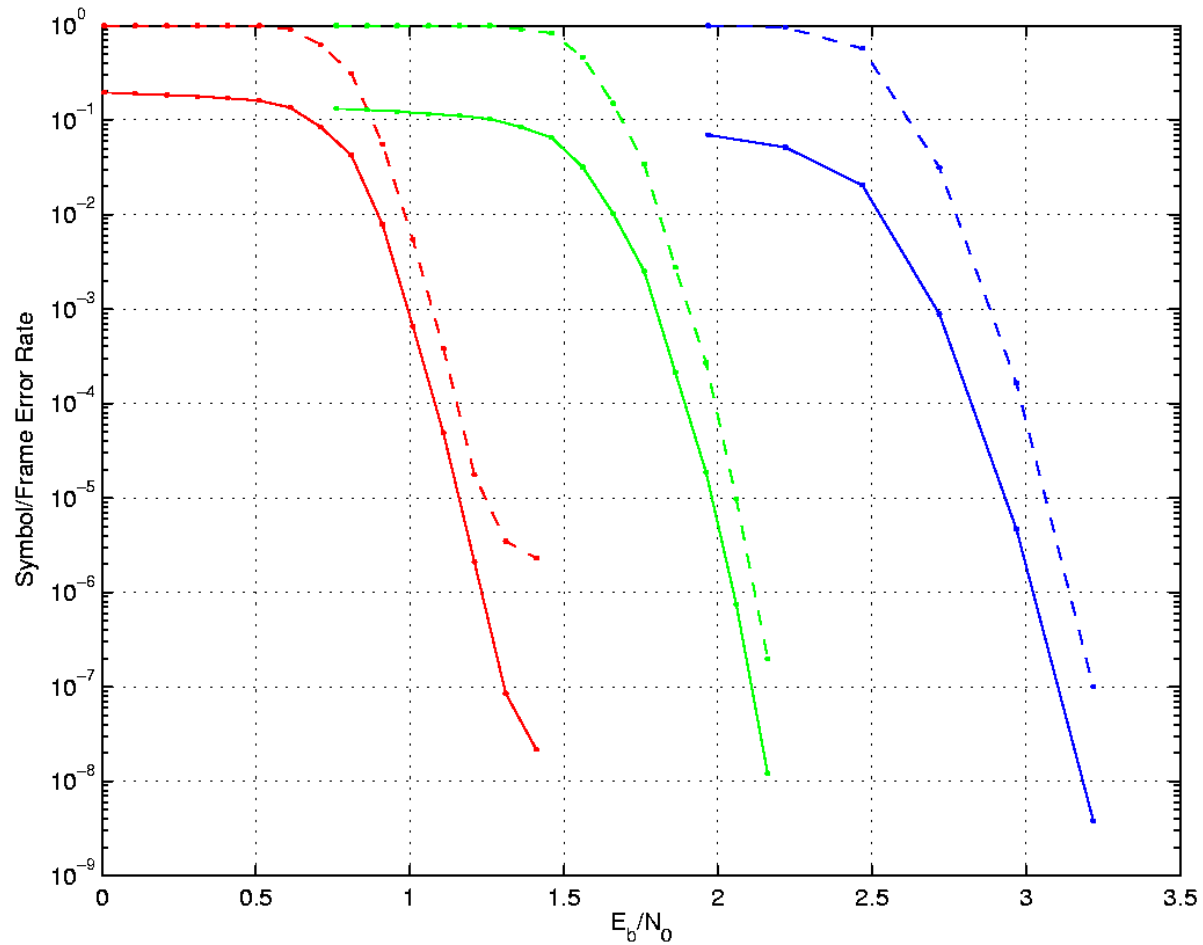
Encoding of message  $m$  requires computing  $mG_{qc}$ . Because  $G_{qc}$  is quasicyclic, this can be performed in an efficient bit-serial manner using  $4n/M$  linear feedback shift registers, each of length  $M/4$ , as shown in figure B-2. Initially, the binary pattern from the first row of circulants is placed in the top row of small boxes in the figure. With the switches set as drawn, the  $k$  message bits are fed through the encoder one at a time, and the registers are updated and shifted once per bit. After each set of  $M/4$  message bits are processed, the circulant patterns are updated for the next row of circulants. Then the switches are changed and the contents of the registers are read out sequentially as the parity portion of the codeword.

## B2 PERFORMANCE

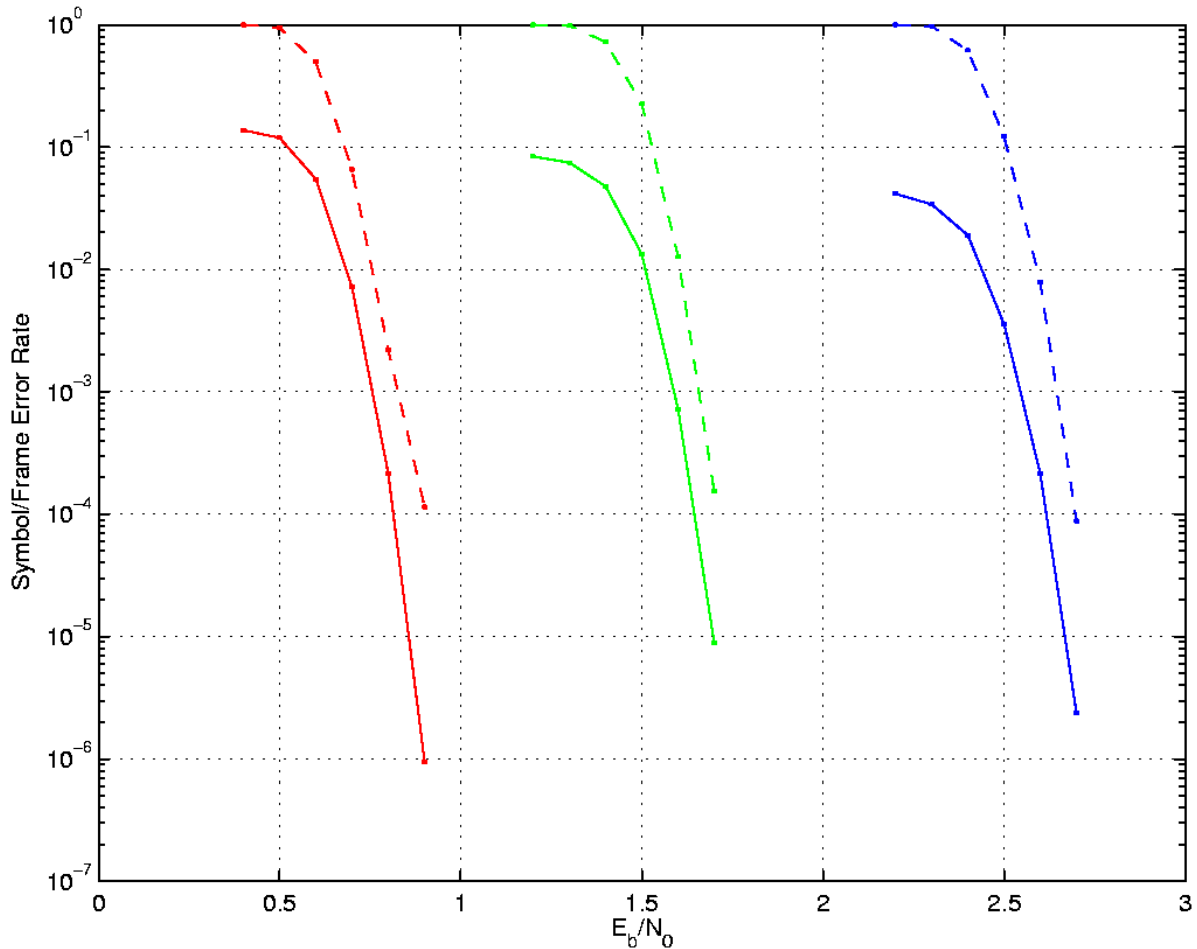
Figure B-3 shows the frame error rates (dashed) and symbol error rates (solid) for the short blocklength members of the code family. From left to right, these three codes have parameters  $(n=2048, k=1024)$  rate  $1/2$ ,  $(n=1536, k=1024)$  rate  $2/3$ , and  $(n=1280, k=1024)$  rate  $4/5$ . Similarly, figure B-4 shows performance curves for the midsize blocklength codes with parameters  $(n=8192, k=4096)$  rate  $1/2$ ,  $(n=6144, k=4096)$  rate  $2/3$ , and  $(n=5120, k=4096)$  rate  $4/5$ . Figure B-5 shows performance curves for the long blocklength codes with parameters  $(n=32768, k=16384)$  rate  $1/2$ ,  $(n=24576, k=16384)$  rate  $2/3$ , and  $(n=20480, k=16384)$  rate  $4/5$ .



**Figure B-3: Performance of Length  $k=1024$  LDPC Codes: Rate  $1/2$ ,  $2/3$ ,  $4/5$  (Left to Right)**



**Figure B-4: Performance of Length  $k=4096$  LDPC Codes: Rate 1/2, 2/3, 4/5 (Left to Right)**



**Figure B-5: Performance of Length  $k=16384$  LDPC Codes: Rate 1/2, 2/3, 4/5 (Left to Right)**

Performance curves for the codes with  $k=1024$  and  $k=4096$  were determined by hardware simulation on a Xilinx Virtex-II FPGA (see reference [C2.5]); performance for the  $k=16384$  codes are from software simulations. In each case, a large maximum number of iterations was allowed, and a stopping rule was used so the average number of iterations required remained small.

It is well known that error floors can be caused either by the structure of the code, or by implementation details of the decoder. Except for the  $(n=1280, k=1024)$  rate 4/5 code, experiments indicate that each code has a minimum distance of at least 30. The error floors that appear in a few cases may be due in part to characteristics of the decoders used.

## ANNEX C

### INFORMATIVE REFERENCES

#### C1 INFORMATIVE REFERENCES FOR SECTION 2, LOW DENSITY PARITY CHECK CODE OPTIMIZED FOR NEAR EARTH APPLICATIONS

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- [C1.4] Y. Kou, S. Lin, and M. P. C. Fossorier. "Low-Density Parity-Check Codes Based on Finite Geometries: A Rediscovery and New Results." *IEEE Trans. Information Theory* 47 (November 2001): 2711-2736.
- [C1.5] W. Fong, "White Paper for Low Density Parity Check (LDPC) Codes for CCSDS Channel Coding Blue Book." CCSDS P1B Channel Coding Meeting, Houston, TX, October 2002.
- [C1.6] Z. Li, et al. "Efficient Encoding of Quasi-Cyclic Low Density Parity Check Codes." *IEEE Transactions on Communication* [to be published].
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- [C1.8] S. Lin and D. Costello, Jr. *Error Control Coding*. 2nd ed. New Jersey: Pearson Prentice Hall, 2004.

## **C2 INFORMATIVE REFERENCES FOR SECTION 3, LOW DENSITY PARITY CHECK CODE FAMILY OPTIMIZED FOR DEEP SPACE APPLICATIONS**

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<[http://tmo.jpl.nasa.gov/progress\\_report/42-159/159K.pdf](http://tmo.jpl.nasa.gov/progress_report/42-159/159K.pdf)>
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<[http://tmo.jpl.nasa.gov/progress\\_report/42-159/159K.pdf](http://tmo.jpl.nasa.gov/progress_report/42-159/159K.pdf)>
- [C2.5] C. Jones, et al. "Approximate-MIN\* Constraint Node Updating for LDPC Code Decoding." In *Proceedings of MILCOM 2003 (Boston, Massachusetts)*, 1-157-1-162. Piscataway, NJ: IEEE, October 2003.
- [C2.6] J. Lee and J. Thorpe. "Memory-Efficient Decoding of LDPC Codes." In *Proceedings of the IEEE International Symposium on Information Theory (Adelaide, Australia)*, 459-463. Piscataway, NJ: IEEE, September 2005.
- [C2.7] *TM Space Data Link Protocol*. Recommendation for Space Data System Standards, CCSDS 132.0-B-1. Blue Book. Issue 1. Washington, D.C.: CCSDS, September 2003.
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NOTE – Normative references are contained in 1.4.